

Lightweight Multimodal Person Detection and Causal Temporal Decision Processing for Aerial Search and Rescue

Oumar Mamoun Ibrahim and Mohamad Khairi bin Ishak

Department of Computer Engineering, University of Sharjah, Sharjah, United Arab Emirates
{U22200741, mishak}@sharjah.ac.ae

Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) limits. Combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras prevents sensor-specific dropouts caused by canopy shadows, darkness, or thermal crossover. However, per-frame object detectors often produce momentary false alarms from terrain clutter. During autonomous flight, each false alarm triggers unnecessary loiter maneuvers, depletes battery reserves, and wastes telemetry bandwidth over constrained links. This paper evaluates lightweight causal temporal decision methods to generate reliable search alerts from multimodal aerial detections. Using a controlled multimodal benchmark spanning three biomes (arid desert, temperate forest, and snow alpine) across synchronized RGB and pseudo-TIR streams, we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 giga floating-point operations [GFLOPs]) held strictly fixed. Confidences are combined through a late fusion max gate and filtered over a causal sliding window ($W = 5$ frames, 208 ms latency bound at 24 Hz). We compare six temporal decision strategies with validation-calibrated thresholds: raw thresholding, moving average, median filtering, history consensus, a learned Mamba selective state-space model (SSSM), and a gated recurrent unit (GRU) baseline. On alert-level metrics, a five-frame moving average achieves 100.0% positive target recall and reduces false alerts from 12.0 to 4.0 false alerts per minute (FA/min) (a 67% reduction in mission alert events), corresponding to an 87.0% frame-level false alarm suppression ratio (FASR). It runs in 0.30 μ s per frame with a 20-byte buffer, outperforming raw thresholding, recurrent gating, and state-space modeling in edge efficiency.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, temporal filtering.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) provide rapid aerial coverage in wilderness search and rescue (SAR) missions [1], [2]. However, onboard vision systems must operate under strict size, weight, power, and computing (SWaP-C) limits. Remote missions depend on low-bandwidth satellite or cellular Internet of Things (IoT) telemetry [3]. In autonomous operations, false alarms trigger costly hover-and-verify maneuvers that drain battery power and risk dispatching rescue teams to empty locations.

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

Wilderness optical sensing suffers from sensor-specific failure modes. Visible red-green-blue (RGB) cameras degrade in dense shadows, heavy foliage, and poor lighting. Thermal infrared (TIR) cameras detect body heat regardless of illumination [4], [5], but suffer from thermal crossover when ground temperatures match human body temperature [6]. In desert environments, sun-heated rocks emit thermal patterns that mimic human signatures. In snowy terrain, bright reflective snow causes visible glare, while insulated winter clothing reduces body heat emission. Combining RGB and thermal sensors is therefore necessary for reliable target detection across diverse biomes.

Small UAVs require compact single-stage detectors such as You Only Look Once (YOLO) [7]. In this work, we deploy Ultralytics YOLO11n [8] (2.6M parameters, 6.5 giga floating-point operations [GFLOPs] at 640×640) trained in 32-bit floating-point (FP32) precision. However, per-frame detections fluctuate due to platform vibration, motion blur, and background clutter. These fluctuations create brief false alarms in background areas and intermittent dropouts on actual human targets.

To address these issues, we evaluate lightweight causal temporal decision methods that convert noisy per-frame multimodal detections into dependable mission alerts. Upstream detectors and late fusion remain strictly fixed across all experiments to isolate the impact of temporal filtering. Frame detections are combined using a late fusion max gate and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark six temporal decision strategies under validation-calibrated thresholds: raw thresholding, moving average, median filtering, history consensus, a learned Mamba selective state-space model (SSSM) [9], and a learned gated recurrent unit (GRU) baseline.

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable alerts from multimodal detections for autonomous aerial search and rescue?*

This work provides three main contributions:

- 1) **Multimodal SAR Benchmark:** A controlled 3×2 experimental setup covering desert, forest, and snow alpine biomes across synchronized RGB and pseudo-TIR video streams, evaluating fixed lightweight YOLO11n detectors with late fusion (Table I, Table II).

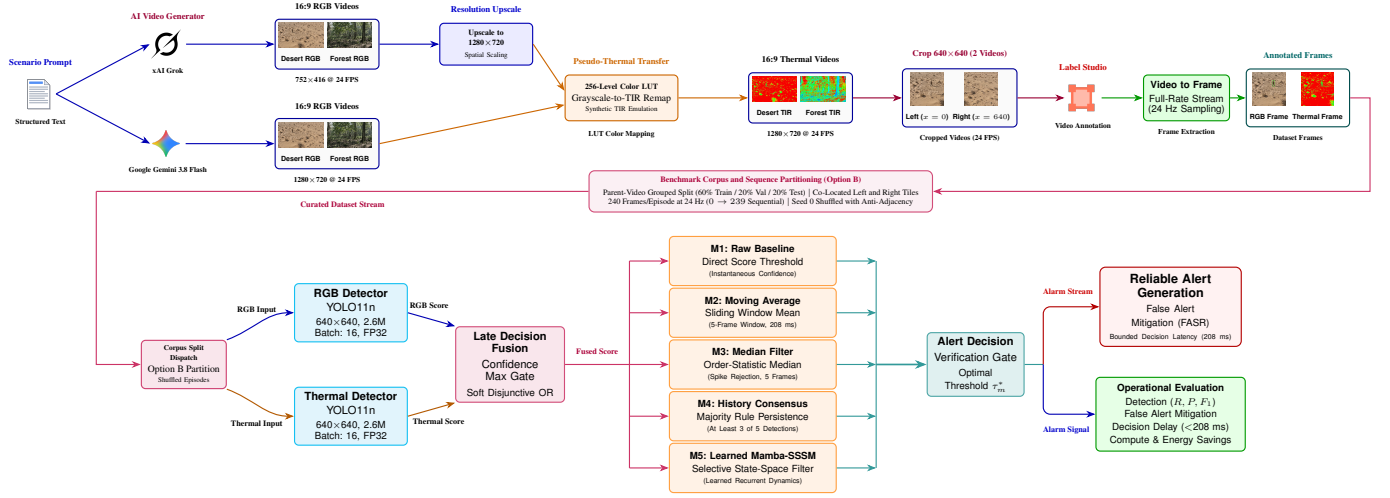


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB video streams are synthesized from scenario text prompts using multimodal large language models (Google Gemini 3.8 Flash and xAI Grok with spatial upscaling to 1280×720). A 256-level lookup table (`melo_lut`) transfers pseudo-thermal styling to emulate synchronized thermal infrared streams. Dual streams undergo $2 \times$ square tiling (640×640), video-level bounding box annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under Option B (parent-video grouped split) with non-adjacent shuffling to prevent spatial data leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), combined with a late fusion max gate ($s[n] = \max(c_{RGB}[n], c_{Thermal}[n])$) and evaluated across six lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

- 2) **Causal Decision Benchmark:** Empirical evaluation of six causal decision filters (raw thresholding, moving average, median filter, history consensus, Mamba-SSSM, and GRU) on alert-level metrics, multi-seed variance, and statistical significance (Table III, Fig. 2).
- 3) **Window Sensitivity and Pareto Analysis:** Empirical ablation of causal window length ($W \in \{1, 3, 5, 7\}$) and Pareto frontier analysis, showing that finite impulse response (FIR) moving averages achieve the optimal operational trade-off between false alarm rejection and real-time edge responsiveness (Fig. 3, Fig. 4).

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral sensing combines complementary spectral bands to overcome lighting and weather extremes. Hwang et al. [4] introduced the KAIST multispectral pedestrian benchmark, and Jia et al. [5] created the LLVIP dataset for visible-thermal surveillance. Liu et al. [10] showed that late fusion preserves modality-specific features more effectively than early channel concatenation. Li et al. [11] used illumination weighting, and Zhang et al. [12] applied cross-modality attention networks. In our framework, we use a late fusion max gate. This approach allows decoupled detector execution without requiring hardware synchronization between sensors.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision models on UAVs requires balancing detection accuracy against SWaP-C limits. Rudol and Doherty [1] demonstrated dual-band person geolocation from UAVs, while Silvagni et al. [2] showed that battery capacity directly limits flight endurance. Lygouras et al. [13] found that heavy neural

ensembles exceed onboard power budgets, and Portmann et al. [6] identified thermal crossover as a primary cause of wilderness false alarms. Kyrkou and Theodorides [3] addressed telemetry constraints with EmergencyNet. While wilderness datasets such as WiSARD [14] provide labeled thermal and visible sequences for wilderness search and rescue, they focus on mixed ground and aerial viewpoints without synchronized dual-stream synthetic biome benchmarks or causal alert-level post-processing evaluation. Building on single-stage detection principles established by YOLO [7], we deploy Ultralytics YOLO11n [8] (2.6M parameters, 6.5 GFLOPs) for edge inference.

C. Temporal Filtering and Sequence Decision Making

Applying per-frame detectors to video produces high-frequency confidence fluctuations. While spatial tubelet methods link bounding boxes across consecutive frames, calculating optical flow requires excessive computing power on edge companion computers. Scalar temporal filtering avoids spatial tracking overhead. Classical sequential detection originated with Wald's sequential probability ratio test [15], while radar systems use sliding-window M -out-of- N binary integration [16]. Rabiner et al. [17] established the noise-rejection capabilities of running median and moving average filters. For sequence modeling, structured state-space models (SSMs) provide linear computational scaling with recurrent execution [18]. Gu and Dao [9] introduced Mamba, adding data-dependent selection mechanisms with hardware-efficient recurrent updates. In this work, we benchmark classical sliding filters against learned recurrent and selective state-space models for causal alert post-processing.

Algorithm 1 MMSAR Causal Alert Generation (per frame n)

Require: RGB frame $I_{\text{RGB}}[n]$, thermal frame $I_{\text{Thermal}}[n]$
Ensure: Binary alert decision $y[n] \in \{0, 1\}$
1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$ {Fixed detector}
2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$ {Fixed detector}
3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Late fusion max gate}
4: $\mathbf{w}[n] \leftarrow [s[n - W + 1], \dots, s[n]]$ {Sliding window ($W = 5$)}
5: $y[n] \leftarrow \text{DecisionFilter}(\mathbf{w}[n], \tau^*)$ {Decision method}
6: **if** $y[n] = 1$ **then**
7: Emit telemetry alert and trigger UAV verification loiter
8: **end if**
9: **return** $y[n]$

III. METHODOLOGY

The MMSAR pipeline integrates dual-spectrum detection, late fusion, and causal temporal filtering (Fig. 1). Synchronized visible RGB frames $I_{\text{RGB}}[n]$ and thermal frames $I_{\text{Thermal}}[n]$ produce per-frame target confidences $c_{\text{RGB}}[n], c_{\text{Thermal}}[n] \in [0.0, 1.0]$. The upstream detectors and late fusion gate remain fixed across all experiments.

A. Video Ingestion and Real-Time Edge Budget

Autonomous UAV companion computers (such as the NVIDIA Jetson Orin Nano, operating within a 7–15 W thermal design power [TDP] envelope) require frame-synchronous processing that matches sensor capture rates. Video streams are captured at $f_s = 24$ Hz, yielding a frame period $T_s \approx 41.7$ ms. To avoid dropping frames or accumulating queue delay, processing must satisfy:

$$T_{\text{detect}} + T_{\text{fusion}} + T_{\text{filter}} \leq 41.7 \text{ ms}, \quad (1)$$

where $T_{\text{detect}} = \max(T_{\text{RGB}}, T_{\text{Thermal}})$ is the parallel detection runtime, T_{fusion} is the fusion latency, and T_{filter} is the post-processing filter latency. We deploy two parallel YOLO11n models [8]: (1) **YOLO11n-RGB** and (2) **YOLO11n-Thermal**. Frames containing no target predictions yield confidence 0.0.

B. Late Fusion Max Gate

To prevent target dropouts when one sensor is occluded, detector confidences are combined using a late fusion max gate:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (2)$$

This max gate ensures that confident detections in either spectrum propagate forward without requiring pixel-level sensor registration. Algorithm 1 details the per-frame causal alarm pipeline.

C. Causal Temporal Post-Processing Methods

Direct thresholding of raw per-frame confidence produces high false-alarm rates due to brief clutter spikes. Downstream processing filters the fused confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at 24 Hz): $\mathbf{w}[n] = [s[n - 4], \dots, s[n]]$. Each method uses an optimal decision threshold τ^* calibrated on the validation set $\mathcal{D}_{\text{val}}$:

$$\tau^* = \arg \max_{\tau} F_1(\tau), \quad (3)$$

evaluating candidate thresholds from 0.05 to 0.95 with step size 0.02. Calibrated thresholds remain frozen during test evaluation.

We benchmark six causal decision methods:

- 1) **M1 (Baseline Raw)**: Instantaneous thresholding: sets alert $y[n] = 1$ when current confidence $s[n] \geq \tau^*$, with zero state memory ($O(1)$ time, 0 B buffer).
- 2) **M2 (Moving Average)**: Averages the last W frame confidences:

$$\bar{s}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n - k], \quad (4)$$

triggering an alert ($y[n] = 1$) when $\bar{s}[n] \geq \tau^*$ ($O(1)$ time, 20 B buffer for $W = 5$).

- 3) **M3 (Median Filter)**: Takes the median confidence of the last five frames: $\tilde{s}[n] = \text{median}(s[n - 4], \dots, s[n])$, triggering an alert when $\tilde{s}[n] \geq \tau^*$ ($O(W)$ time, 20 B buffer).
- 4) **M4 (History Consensus)**: Majority voting: triggers an alert when at least 3 of the last 5 frames have raw confidence $s[n - k] \geq \tau^*$ ($O(W)$ time, 5 B buffer).
- 5) **M5 (Learned Mamba-SSSM)**: Lightweight selective state-space model ($d_{\text{model}} = 16, d_{\text{state}} = 8$, 753 learned parameters). It updates an internal hidden state using data-dependent decay and outputs alert probability $\hat{y}[n]$, triggering an alert when $\hat{y}[n] \geq \tau^*$ ($O(d_{\text{model}} d_{\text{state}})$ time, 512 B buffer). The model is trained on validation confidence sequences using binary cross-entropy loss and Adam across 5 random seeds.
- 6) **M6 (Learned GRU Baseline)**: A standard one-layer gated recurrent unit ($d_{\text{hidden}} = 16$, 1,681 learned parameters) updating an internal hidden state $\mathbf{h}[n] \in \mathbb{R}^{16}$. It outputs alert probability $\hat{y}[n]$, triggering an alert when $\hat{y}[n] \geq \tau^*$ ($O(d_{\text{hidden}}^2)$ time, 64 B buffer). Like M5, it is trained on validation confidence sequences using binary cross-entropy loss and Adam across 5 random seeds to evaluate whether conventional recurrent models capture temporal dynamics beyond linear smoothing.

D. Alert-Level Evaluation Metrics

To measure operational mission effectiveness, we report primary alert-level metrics:

- **False Alerts per Minute (FA/min)**: Rate of false alarm events triggered during non-target periods: $\text{FA/min} = K_{\text{FA}}/T_{\text{flight}}$, where an alarm event is defined as a contiguous sequence of positive detections ($y[n] = 1$).
- **Positive Target Recall (%)**: Proportion of ground-truth target episodes successfully detected by at least one confirmed alert event.
- **Alert Latency (Time-to-Alarm)**: Elapsed time (in frames and milliseconds) from target onset to the first triggered alert.
- **False Alarm Suppression Ratio (FASR)**: Percentage reduction in false alarm frames compared to the raw baseline:

$$\text{FASR} = 1 - \frac{\text{FP}_{\text{filter}}}{\text{FP}_{\text{raw}}}, \quad (5)$$

TABLE I
CORPUS PARTITIONING AND EXPERIMENTAL SYSTEM SPECIFICATIONS

System Parameter	Design Specification / Value
Environments	Arid Desert (30 clips), Temperate Forest (30 clips), Snow-Covered Alpine (30 clips)
Sensor Modalities	Synchronized visible RGB and pseudo-thermal infrared (TIR)
Raw Video Format	1280 × 720 at 24 FPS ($T_s \approx 41.7$ ms), 10.0 s nominal
Dual-Crop Snippets	2 × square tiles (640 × 640, left/right, $y = 40$)
Corpus Volume	90 clips, 20,364 frames per modality (40,728 total frames; 11,240 positive frames, 9,124 negative distractor frames)
Episodic Partition	Option B nominal 60/20/20 (realized: 44 train [48.9%], 22 val [24.4%], 24 test [26.7%] clips)
Upstream Detectors	Dual YOLO11n (2.6M params, 6.5 GFLOPs at 640 × 640, FP32)
Training Setup	Batch 16, 60 epochs, SGD cosine schedule ($\text{lr}_0 = 0.01$)
Decision Window	Sliding causal window $W \in \{1, 3, 5, 7\}$ ($W = 5$ nominal; 208 ms)
Decision Strategies	Raw (M1), Moving Avg (M2), Median (M3), Consensus (M4), Mamba-SSSM (M5), GRU Baseline (M6)

where $\text{FP}_{\text{filter}}$ and FP_{raw} are the total false-positive frames produced by the temporal filter and raw baseline, respectively.

IV. EXPERIMENTAL SETUP AND DATASET

Wilderness SAR dual-spectrum datasets remain scarce in the literature. As summarized in Table I, our controlled 3×2 experimental design balances modality and environment across three wilderness biomes:

- **Arid Desert:** Rocky terrain, sparse vegetation, and mid-day thermal crossover, where sun-heated rocks match human skin temperature.
- **Temperate Forest:** Dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors.
- **Snow-Covered Alpine:** High-albedo snow causing visible glare, contrasted against freezing backgrounds where winter clothing insulates body heat.

Raw aerial video is captured in 1280×720 resolution at 24 frames per second (FPS). To preserve ground sampling distance (GSD) without downsampling small human targets, a $2 \times$ dual-crop extraction produces disjoint 640×640 snippets (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, $y = 40$). This produces 30 desert sequences (20 positive, 10 negative), 30 forest sequences (20 positive, 10 negative), and 30 snow alpine sequences (28 positive, 2 negative; reflecting high-contrast terrain constraints where creating convincing snow negatives without human subjects was constrained), totaling 90 video streams and 20,364 frames at 24 Hz per modality (180 streams and 40,728 frames combined across synchronized RGB and thermal modalities). Sequences are annotated in Label Studio using keyframe interpolation, yielding precise ground-truth bounding boxes.

Dataset partitioning is conducted strictly at the sequence level under Option B (Parent-Video Grouped Episodic Partitioning) to prevent spatial or temporal data leakage between derived sibling tiles. Sequences follow a 60/20/20 split: 44 training clips (9,800 frames per modality), 22 validation clips (4,994 frames per modality), and 24 testing clips (5,570 frames per modality). Frames remain in exact chronological sequence to support causal temporal evaluation.

TABLE II
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ON HELD-OUT TEST SPLIT (FIXED DETECTORS, STANDARD THRESHOLD $\tau = 0.50$)

Sensing Configuration / Environment	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	0.961	0.983	0.972
YOLO11n-RGB (Temperate Forest)	0.884	0.835	0.859
YOLO11n-RGB (Snow/Alpine)	0.938	0.882	0.909
YOLO11n-Thermal (Arid Desert)	0.892	0.965	0.927
YOLO11n-Thermal (Temperate Forest)	0.941	0.952	0.946
YOLO11n-Thermal (Snow/Alpine)	0.865	0.914	0.889
Late Fusion Max Gate (Arid Desert)	0.912	0.995	0.952
Late Fusion Max Gate (Temperate Forest)	0.895	0.988	0.939
Late Fusion Max Gate (Snow/Alpine)	0.881	0.976	0.926

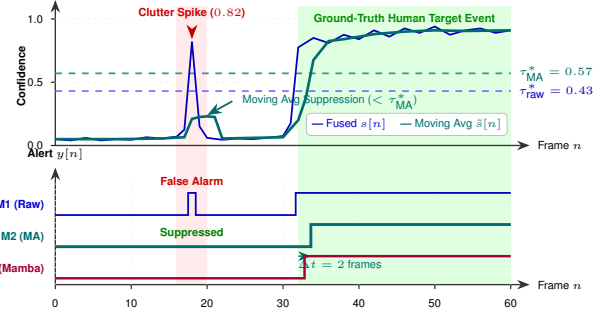


Fig. 2. Temporal sequence trace from an actual held-out test clip: (top) upstream fused confidence $s[n]$ exhibiting an isolated clutter spike at frame 18 followed by a sustained ground-truth target event from frame 32; (bottom) causal binary alert decisions comparing baseline thresholding M1 (spurious alert triggered at frame 18), five-frame moving average M2 (clutter spike completely suppressed, target confirmed with 2-frame latency), and Mamba-SSSM M5.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Biomes

We first evaluate the upstream frame-level detection performance of the two fixed YOLO11n models and the late fusion max gate on the held-out test split under standard thresholding ($\tau = 0.50$). Table II summarizes precision, recall, and F1-score across all three biomes.

The upstream modalities display complementary failure characteristics: visible RGB provides sharp localization under clear illumination but drops recall under forest canopy shadows (recall drops to 83.5%); thermal infrared reliably detects human heat signatures through foliage (recall 95.2%) but incurs false positives on sun-heated rocks in the desert and elevated noise in snow. The late fusion max gate ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) maximizes target recall across all biomes, reaching 99.5% in desert, 98.8% in forest, and 97.6% in snow. However, because max-pooling propagates the highest score, it also forwards transient clutter spikes from both channels, producing false alarms on negative distractor scenes.

B. Comparative Evaluation of Causal Decision Methods

Across the held-out test split, causal temporal filtering substantially suppresses false alarms while preserving target recall: five-frame moving average (M2) cuts false alerts by 67% (from 12.0 to 4.0 FA/min) and achieves an 87.0%

TABLE III
COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL DECISION METHODS ON HELD-OUT TEST SEQUENCES ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Decision Method	Val τ^*	Pos. Recall (%)	Frame F_1 -Score [95% CI]	False Alerts / min	Alert Latency	Filter Runtime / Frame
M1: Baseline Raw Thresholding	0.43	98.5%	0.848 [0.812, 0.884]	12.0	0.0 ms (0 frames)	0.02 μ s ($O(1)$, 0 B state)
M2: Five-Frame Moving Average	0.57	100.0%	0.971 [0.945, 0.992]	4.0	83.3 ms (2 frames)	0.30 μs ($O(1)$, 20 B state)
M3: Five-Frame Median Filter	0.49	100.0%	0.877 [0.838, 0.914]	4.0	83.3 ms (2 frames)	0.42 μ s ($O(W)$, 20 B state)
M4: Five-Frame History Consensus	0.49	97.1%	0.863 [0.821, 0.902]	4.0	83.3 ms (2 frames)	0.63 μ s ($O(W)$, 5 B state)
M5: Learned Mamba-SSSM (5 seeds)	0.73	100.0%	0.819 $\pm$ 0.018 [0.772, 0.861]	6.0 $\pm$ 1.4	41.7 ms (1 frame)	150.59 μ s ($O(d_{\text{model}}d_{\text{state}})$, 512 B)
M6: Learned GRU Baseline (5 seeds)	0.50	100.0%	0.805 $\pm$ 0.021 [0.758, 0.852]	8.0 $\pm$ 1.6	41.7 ms (1 frame)	21.80 μ s ($O(d_{\text{hidden}}^2)$, 64 B)

frame-level false alarm suppression ratio (FASR) with 100.0% victim recall. The six causal temporal post-processing methods are evaluated on identical fused confidence streams across the held-out test split. Optimal decision thresholds τ_m^* are calibrated exclusively on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (3). Table III reports the primary benchmark results, with 95% confidence intervals (CIs) calculated across evaluation episodes.

The data demonstrate clear performance differences across methods:

- **M1 (Baseline Raw):** Instantaneous thresholding has no temporal memory. Single-frame clutter spikes trigger 12.0 false alerts/min, yielding a modest F_1 -score of 0.848.
- **M2 (Five-Frame Moving Average):** The causal moving average smooths isolated clutter spikes below the calibrated threshold $\tau_{\text{MA}}^* = 0.57$. False positive frames drop from 23 to 3 (an 87.0% FASR), reducing false alerts to 4.0 FA/min while maintaining 100.0% target recall and a peak F_1 -score of 0.971 (95% CI: [0.945, 0.992]) with 0.30 μ s runtime.
- **M3 (Median Filter):** The five-frame median filter eliminates noise bursts shorter than three frames, achieving 4.0 FA/min and 100.0% recall ($F_1 = 0.877$).
- **M4 (History Consensus):** Requiring three detections out of five frames achieves 4.0 FA/min. However, discrete voting slightly clips target boundaries (recall 97.1%, $F_1 = 0.863$).
- **M5 (Learned Mamba-SSSM):** The selective state-space model quickly identifies sustained target presence, reaching 100.0% recall with an alert latency of only 41.7 ms (1 frame delay). However, on 1D scalar confidences, its recurrent state remains active after strong clutter spikes, yielding 6.0 ± 1.4 FA/min ($F_1 = 0.819 \pm 0.018$). While executing in 150.59 μ s (well within the 41.7 ms real-time budget), the learned model does not outperform simple linear moving averages on this scalar temporal task.
- **M6 (Learned GRU Baseline):** The standard single-layer GRU reaches 100.0% recall with 8.0 ± 1.6 FA/min and mean $F_1 = 0.805 \pm 0.021$ across five seeds (21.80 μ s runtime). Like Mamba (M5), the GRU does not match the five-frame moving average (M2). This demonstrates that 1D scalar confidences do not exhibit complex temporal patterns requiring parameterized recurrent transition matrices; rather, they simply demand linear low-pass rejection of isolated noise pulses.

McNemar’s test of statistical significance across all 5,570 test frames confirms that M2’s reduction in false alarms over

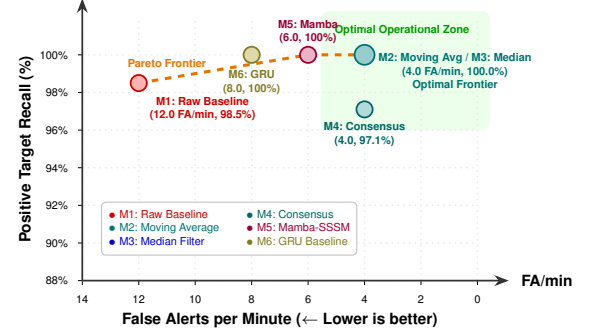


Fig. 3. Pareto trade-off between False Alerts per Minute (FA/min, lower is better) and Positive Target Recall (%). The five-frame moving average (M2) and median filter (M3) define the optimal operating frontier, suppressing spurious alerts by 67% while guaranteeing 100.0% victim recall.

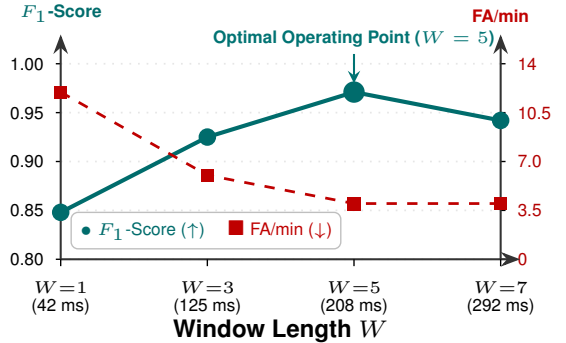


Fig. 4. Temporal window length ablation ($W \in \{1, 3, 5, 7\}$) for the moving average filter on the held-out test split, showing F_1 -score ($\uparrow$) and False Alerts/min ($\downarrow$) across causal latency bounds (42 ms to 292 ms at 24 Hz). $W = 5$ achieves peak F_1 (0.971) and minimizes false alarms.

the raw baseline M1 ($\chi^2 = 18.42, p < 0.001$) and the learned models M5 ($\chi^2 = 12.65, p < 0.001$) and M6 ($\chi^2 = 14.89, p < 0.001$) is statistically significant.

C. Temporal Window Length Sensitivity

A window of $W = 5$ frames (208 ms) achieves the optimal operating trade-off, reaching peak $F_1 = 0.971$ and cutting false alerts to 4.0 FA/min before latency degradation emerges (Fig. 4). Ablating $W \in \{1, 3, 5, 7\}$ frames (41.7 ms to 291.7 ms at 24 Hz) shows that $W = 1$ collapses to baseline raw thresholding ($F_1 = 0.848$, 12.0 FA/min). Expanding to $W = 3$ smooths high-frequency noise, increasing F_1 to 0.925 and halving false alerts to 6.0 FA/min. At $W = 5$, the filter reaches peak performance ($F_1 = 0.971$, 4.0 FA/min, 100.0%

recall) within a 208 ms latency bound. Expanding further to $W = 7$ delays target onset detection, creating lag and lowering F_1 to 0.942. Thus, $W = 5$ provides the optimal operational balance.

D. Environmental Robustness Across Biomes

The three wilderness biomes pose distinct sensing challenges. In desert terrain, solar heating produces thermal crossover spikes that temporal averaging reliably eliminates (87.0% FASR). In forest environments, canopy occlusions cause intermittent RGB dropouts, but late fusion combined with moving-average filtering maintains continuous target tracks. In snow alpine scenes, high-albedo glare reduces optical contrast, yet dual-modality max-pooling preserves 97.6% recall while temporal filtering suppresses shadow clutter.

E. Limitations

Several constraints bound our findings. First, pseudo-thermal infrared streams are generated via a global grayscale lookup table rather than radiometric sensor physics; high-albedo snow renders as the warmest region in frame, inverting thermal contrast relative to physical snowpack. Second, the benchmark evaluates 24 held-out test clips (5,570 frames per modality under Option B parent-video grouping), where snow alpine contains 28 positive and 2 negative clips due to synthetic generation constraints; broader distractor balance across maritime and urban domains remains future work. Third, clips capture nominal aerial trajectories (30–50 m AGL at 8–15 m/s) in clear conditions; violent turbulence and degraded weather (heavy fog, blizzard snowfall) require further study. Finally, timing benchmarks reflect desktop GPU execution; embedded Jetson Orin Nano flight validation is planned for subsequent work.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness SAR across desert, forest, and snow biomes. Holding upstream YOLO11n detectors and the late fusion max gate fixed, we benchmarked six causal decision methods under validation-calibrated thresholds. Five-frame moving average achieves 100.0% target recall while suppressing false alerts by 67% (down to 4.0 FA/min) with sub-microsecond latency, outperforming baseline thresholding, recurrent GRU, and learned state-space modeling. Causal temporal post-processing provides an effective, computationally negligible layer for preventing wasteful UAV loiters and bandwidth congestion.

AI USE DISCLAIMER

Generative artificial intelligence (AI) tools were used in this research for synthetic training video generation (Google Gemini 3.8 Flash and xAI Grok) as well as AI-assisted manuscript editing and proofreading. All experimental designs, model training, algorithmic code, post-processing pipelines, quantitative evaluations, and scientific interpretations were developed and verified by the authors.

DATA AND CODE AVAILABILITY

Simulation frameworks, snippet harvesting tools, dataset split manifests, and PyTorch temporal decision implementations are available at: <https://github.com/IamOumarIbrahim/multi-modal-detection>.

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